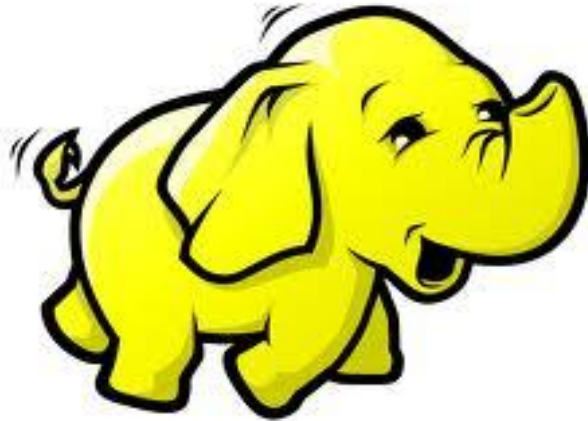




MORE ON HADOOP

- Hadoop Commands
- Monitoring and Managing
- Scaling Out vs Scaling Up
- Hadoop Streaming
- Running Hadoop jobs
- High Availability NameNode

Hadoop Commands

- General syntax:
 - `hadoop command [genericOptions] [commandOptions]`

Hadoop Commands

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- `hadoop command [genericOptions] [commandOptions]`

Command	Description
fs	Executes a File System Command
jar	Runs a jar file
job	Manages the jobs

Hadoop Commands

■ General syntax:

- `hadoop command [genericOptions] [commandOptions]`

Option	Description
<code>-conf <configuration file></code>	Specify an application configuration file.
<code>-D <property>=<value></code>	Use value for given property.
<code>-files <comma separated list of files></code>	Specify comma separated files to be copied to the map reduce cluster. Applies only to job.
<code>-libjars <comma separated list of jars></code>	Specify comma separated jar files to include in the classpath. Applies only to job.
<code>-archives <comma separated list of archives></code>	Specify comma separated archives to be unarchived on the compute machines. Applies only to job.

Commands Option

-ls

- **Usage:**

- **hdfs dfs -ls [-h] [-R] uri or path**
- For a file ls returns stat on the file with the following format:
permissions number_of_replicas userid groupid filesize modification_date modification_time filename
- For a directory it returns list of its direct children as in Unix. A directory is listed as:
permissions userid groupid modification_date modification_time dirname
- Files within a directory are order by filename by default

- **Options:**

- -h: Format file sizes in a human-readable fashion (eg 64.0m instead of 67108864)
- -R: Recursively list subdirectories encountered

- **Example:**

- *hdfs dfs -ls /user/*
- *hdfs dfs -ls -R /user/*
- *hdfs dfs -ls hdfs://localhost/user/root/*

- **Exit Code:**

- Returns 0 on success and -1 on error

Commands Option

-mkdir

- **Usage:**
 - `hdfs dfs -mkdir [-p] <paths>`
 - Takes path uri's as argument and creates directories
- **Options:**
 - The -p option behaviour is much like Unix mkdir -p, creating parent directories along the path
- **Example:**
 - `hdfs dfs -mkdir /user/root/`
 - `hdfs dfs -mkdir -p /user/root/dir1 /user/root/dir2`
 - `hdfs dfs -mkdir -p hdfs://localhost/user/root/hadoop`
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-put

- **Usage:**
 - `hdfs dfs -put <localsrc> <dst>`
 - Copy single src, or multiple srcs from local file system to the destination file system
- **Examples:**
 - `hdfs dfs -put localfile5 /user/root/localfile6`
 - `hdfs dfs -put localfile1 localfile2 /user/root/hadoopfile`
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-get

- **Usage:**

- `hdfs dfs -get <src> <localdst>`
- Copy files to the local file system

- **Examples:**

- `hdfs dfs -get /user/root/localfile3`
- `hdfs dfs -get hdfs://localhost/user/root/hadoopfile localfiles`

- **Exit Code:**

- Returns 0 on success and -1 on error

Commands Option

-copyFromLocal

- **Usage:**
 - **`hdfs dfs -copyFromLocal <localsrc> URI`**
 - Similar to put command, except that the source is restricted to a local file reference
- **Options:**
 - The -f option will overwrite the destination if it already exists
- **Examples:**
 - `hdfs dfs -copyFromLocal ./localfile3`
 - `hdfs dfs -copyFromLocal ./localfile3 /user/root/`
 - `hdfs dfs -copyFromLocal file:///root/localfile3 hdfs://localhost/user/root/`
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-copyToLocal

- **Usage:**
 - **`hdfs dfs -copyToLocal URI <localdst>`**
 - Similar to get command, except that the destination is restricted to a local file reference
- **Examples:**
 - *`hdfs dfs -copyToLocal /user/root/localfile3`*
 - *`hdfs dfs -copyToLocal hdfs://localhost/user/root/localfile3`*
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-cat

- **Usage:**
 - `hdfs dfs -cat URI or path`
 - Copies source paths to stdout
- **Examples:**
 - `hdfs dfs -cat hdfs://localhost/user/root/localfile3`
 - `hdfs dfs -cat file:///root/localfile2 /user/root/localfile4`
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-cp

- **Usage:**
 - **`hdfs dfs -cp [-f] URI or path <dest>`**
 - Copy files from source to destination.
 - This command allows multiple sources as well in which case the destination must be a directory
- **Examples:**
 - *`hdfs dfs -cp /user/root/localfile1 /user/root/localfile4`*
 - *`hdfs dfs -cp /user/root/localfile1 /user/root/localfile2 /user/root/dir1`*
 - *`hdfs dfs -cp hdfs://localhost/user/root/localfile1
hdfs://localhost/user/root/hadoopfile4`*
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-df

- **Usage:**
 - `hdfs dfs -df [-h] URI or path`
 - Displays free space
- **Options:**
 - The -h option will format file sizes in a “human-readable” fashion (e.g 64.0m instead of 67108864)
- **Example:**
 - `hdfs dfs -df -h /user/root`

Commands Option

-du

- **Usage:**

- `hdfs dfs -du [-s] [-h] URI or path [URI ...]`
- Displays sizes of files and directories contained in the given directory or the length of a file in case its just a file

- **Options:**

- The -s option will result in an aggregate summary of file lengths being displayed, rather than the individual files.
- The -h option will format file sizes in a “human-readable” fashion (e.g 64.0m instead of 67108864)

- **Example:**

- `hdfs dfs -du /user/root/`
- `hdfs dfs -du -s hdfs://localhost/user/root/`

- **Exit code:**

- Returns 0 on success and -1 on error

Commands Option

-moveFromLocal, -moveToLocal

- **Command:** moveFromLocal
 - **Usage:** `hdfs dfs -moveFromLocal <localsrc> <dst>`
 - Similar to put command, except that the source local src is deleted after it's copied
 - **Example:** `hdfs dfs -moveFromLocal localfile /user/root/`
- **Command:** moveToLocal
 - **Usage:** `hdfs dfs -moveToLocal <src> <dst>`
 - Displays a “Not implemented yet” message

Commands Option

-mv

- **Usage:**
 - `hdfs dfs -mv URI [URI ...] <dest>`
 - Moves files from source to destination
 - Allows multiple sources as well in which case the destination needs to be a directory
 - Moving files across file systems is not permitted
- **Example:**
 - `hdfs dfs -mv /user/root/localfile1 /user/root/localfile5`
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-rm

- **Usage:**
 - `hdfs dfs -rm [-f] [-r | -R] URI or path [URI ...]`
 - Delete files specified as args
- **Options:**
 - The -f option will not display a diagnostic message or modify the exit status to reflect an error if the file does not exist
 - The -R option deletes the directory and any content under it recursively.
 - The -r option is equivalent to -R
- **Example:**
hdfs dfs -rm hdfs://localhost/user/root/localfile /user/localhost/user/root/emptydir
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-tail

- **Usage:**
 - **hdfs dfs -tail [-f] URI or path**
 - Displays last kilobyte of the file to stdout
- **Options:**
 - The -f option will output appended data as the file grows, as in Unix
- **Example:**
 - *hdfs dfs -tail pathname*
- **Exit Code:**
 - Returns 0 on success and -1 on error

Commands Option

-text

- **Usage:**
 - **`hdfs dfs -text <src>`**
 - Takes a source file and outputs the file in text format
- **Example:**
 - *`hdfs dfs -text /user/root/filename`*

Commands Option

-test

- **Usage:**
 - **hdfs dfs -test -[defz] URI**
 - Tests file/directory properties
- **Options:**
 - -d: if the path is a directory, return 0.
 - -e: if the path exists, return 0.
 - -f: if the path is a file, return 0.
 - -s return 0 if file <path> is greater than zero bytes in size
 - -z: if the file is zero length, return 0.
- **Example:**
 - *hdfs dfs -test -e pathname*

Commands Option

-touchz

- **Usage:**
 - **hdfs dfs -touchz URI [URI ...]**
 - Create a file of zero length
- **Example:**
 - *hdfs dfs -touchz pathname*
- **Exit code:**
 - Returns 0 on success and -1 on error

Commands Option -appendToFile

- **Usage:**

- **`hdfs dfs -appendToFile <localsrc> <dst>`**
- Append single src, or multiple srcs from local file system to the destination file system
- Also reads input from stdin and appends to destination file system

- **Examples:**

- *`hdfs dfs -appendToFile localfile /user/hadoop/localfile2`*
- *`hdfs dfs -appendToFile localfile1 localfile2 /user/hadoop/hadoopfile`*
- *`hdfs dfs -appendToFile localfile hdfs://localhost/hadoop/localfile2`*

Commands Option -checksum

- **Usage:**
 - **`hdfs dfs -checksum URI`**
 - Returns the checksum information of a file
- **Example:**
 - *`hdfs dfs -checksum hdfs://localhost/file1`*

Commands Option

-setrep

- **Usage:**
 - `hdfs dfs -setrep [-R] [-w] <numReplicas> <path>`
 - Changes the replication factor of a file
 - If *path* is a directory then the command recursively changes the replication factor of all files under the directory tree rooted at *path*
- **Options:**
 - The `-w` flag requests that the command wait for the replication to complete. This can potentially take a very long time.
 - The `-R` flag is accepted for backwards compatibility. It has no effect.
- **Example:**
 - `hdfs dfs -setrep -w 3 /user/root/dir1`
- **Exit code:**
 - Returns 0 on success and -1 on error

Commands Option

-stat

- **Usage:**

- **hdfs dfs -stat [format] <path> ...**
- Print statistics about the file/directory at <path> in the specified format.
- Format accepts filesize in blocks (%b), type (%F), group name of owner (%g), name (%n), block size (%o), replication (%r), user name of owner(%u), and modification date (%y, %Y). %y shows UTC date as “yyyy-MM-dd HH:mm:ss” and %Y shows milliseconds since January 1, 1970 UTC.
- If the format is not specified, %y is used by default

- **Example:**

- *hdfs dfs -stat "%F %u:%g %b %y %r %n" /file*

- **Exit code:**

- Returns 0 on success and -1 on error

Commands Option

-count

- **Usage:**
 - **`hdfs dfs -count <paths>`**
 - Count the number of directories, files and bytes under the paths that match the specified file pattern
 - The output columns with `-count` are: DIR_COUNT, FILE_COUNT, CONTENT_SIZE, PATHNAME
- **Example:**
 - *`hdfs dfs -count hdfs://localhost/file1 hdfs://localhost/file2`*
- **Exit code:**
 - Returns 0 on success and -1 on error

Commands Option

-getmerge

- **Usage:**

- **`hdfs dfs -getmerge <src> <localdst>`**
- Takes a source directory and a destination file as input and concatenates files in src into the destination local file
- Optionally `-addnl` can be set to enable adding a newline character at the end of each file

- **Example:**

- *`hdfs dfs -getmerge /user/root/temp localfile`*

Commands Option

-help

- Usage
 - `hdfs dfs -help <command>`
 - Returns usage output
- Example
 - *`hdfs dfs -help`*
 - *`hdfs dfs -help ls`*

Managing and Monitoring

HDFS Commands

Checking System's health

- **Command: fsck**
 - Check the health of all the files under a path
 - Reports on each file in HDFS

— Example

- *hdfs fsck /*
- *hdfs fsck -blocks -files -locations*

HDFS Commands

Checking DataNodes health

- **Command: dfsadmin**
 - For reporting on each DataNode

— Example

- *hdfs dfsadmin -report*

HDFS Commands

Entering/Leaving safemode

- **Command: safemode**
 - *hdfs dfsadmin -safemode [enter|leave]*

HDFS Commands

Setting Permissions

- **Commands: chmod, chown, chgrp**
 - **Work similar to Unix commands**

— Example

- *hdfs dfs -chmod 777 <filename>*
- *hdfs dfs -chown user:group <filename>*

HDFS Commands

Removing DataNodes

- DataNodes can be removed from HDFS on the fly.
- Set *dfs.hosts.exclude* configuration parameter to point to a file that has the list of datanodes to be removed. (One node per line)
- Execute the following command
 - *hdfs dfsadmin -refreshNodes*

HDFS Commands

Adding DataNodes

- Install Hadoop on the new DataNodes.
- Start the DataNode daemon manually
 - *hdfs datanode*
- It automatically contacts the NameNode and joins the cluster.
- Add the DataNode to *slaves* file.
- To balance the distribution of files in HDFS
 - *start-balancer.sh*

MAPRED Commands

Managing Jobs

- **Command: list**
 - Lists the currently running jobs

— Example

- *hadoop job -list*
or
- *mapred job -list*

MAPRED Commands

Managing Jobs

- **Command: status**
 - Reports the status of a job

— Example

- *hadoop job -status <job-id>*
or
- *mapred job -status <job-id>*

Job Command Options

COMMAND_OPTION	Description
<code>-submit <job-file></code>	Submits the job.
<code>-status <job-id></code>	Prints the map and reduce completion percentage and all job counters.
<code>-kill <job-id></code>	Kills the job.
<code>-events <job-id> <from-event-#> <#-of-events></code>	Prints the events' details received by jobtracker for the given range.
<code>-history [all] <jobOutputDir></code>	Prints job details, failed and killed tip details. [all] displays more details about the job such as successful tasks and task attempts made for each task.
<code>-list [all]</code>	displays only jobs which are yet to complete. [all] displays all jobs.
<code>-kill-task <task-id></code>	Kills the task. Killed tasks are NOT counted against failed attempts.
<code>-fail-task <task-id></code>	Fails the task. Failed tasks are counted against failed attempts.
<code>-set-priority <job-id> <priority></code>	Changes the priority of the job. Allowed priority values are VERY_HIGH, HIGH, NORMAL, LOW, VERY_LOW

Thank You